ARMORER'S BENCH TECHNICAL MANUAL

Ruger Mark I–IV Target Pistols

Ruger • Mark I, II, III, IV & 22/45 Target & Hunter Models (.22 LR)

Manufacturer:	Ruger	**Category:**	Pistols
Action Type:	Blowback Rimfire Target Pistol	**Era / Service:**	American Rimfire Legend (1949–Present)
Calibers:	.22 LR (Long Rifle)	**Covered Models:**	Standard, Mark I, Mark II Target, Mark III Hunter, Mark IV 22/45, Mark IV Target

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Mark IV One-Button Takedown Pivot • Mark I-III Hammer Strut Alignment • Loaded Chamber Indicator Removal (Mk III) • Volquartsen Accurizing Kit Install • Extractor Claw Upgrade

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
OPERATING SYSTEM	Straight blowback cylindrical bolt cycling inside fixed tubular steel receiver
MARK IV TAKEDOWN	Single rear recessed push-button releases upper receiver from frame on forward hinge pin
MARK II RE■■ASSEMBLY RULE	Muzzle must point upward 45° so hammer strut drops freely into mainspring housing cup before latching
TRIGGER PULL WEIGHT	3.5 to 4.5 lbs (factory) / 1.75 to 2.25 lbs (with Volquartsen target sear and hammer)

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Mark II / III Reassembly Jam (Mainspring Housing Won't Close) **Armorer attempts to close rear mainspring latch, but it meets solid mechanical wall 1/2" before closing**	• Hammer strut is dangling outside the mainspring housing cup. • Hammer not pulled fully forward into uncocked position before swinging housing.	**Immediate Action: Point muzzle 45° straight up. Pull trigger to ensure hammer is fully forward. Listen for hammer strut click. Tilt pistol back so strut seats into the center notch of the mainspring housing plunger. Close latch smoothly.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Ruger Mark IV Detail Disassembly & Volquartsen Trigger Job	Brownells Gunsmithing Bench	Complete walk-through of the Mark IV one-button takedown, stripping the bolt, replacing the extractor with an EDM tool-steel claw, and installing target sears.